Joshi's tasks v2: compare errors with screenshots

Replaces the old tasks (B1-B3, email, paper table) · Due Fri 16 Oct · about 2 hours

In one line. Watch 10 short videos, take a screenshot at the biggest moment of the movement, and tick the answers. No long writing, no coding. Your screenshots will be used in the team presentation and the paper figure, **with your name**.

Why we need this (1 minute)

- A teammate built a computer model. It looks at a person's skeleton data and scores whether an exercise is done right or wrong. It works fairly well, but there are **two things the computer cannot explain**.
- **Question 1 (CTK, hiding face)**: errors E1 and E3 have the same description ("arms not raised enough"). But the model finds E1 well and almost misses E3. Is E3 really a different movement?
- **Question 2 (ELK, flank stretch)**: the model finds errors well for people P1 and P3 but not for P2. What is different about P2?
- A human eye can answer these questions quickly. That is why we need you.

Getting ready (10 minutes)

- Get the **Joshi_videos** folder (163 MB, 10 videos) from the team lead by USB or shared folder.
- Open the videos with **VLC** (free). Space = pause, E = next frame, **Shift+S = screenshot** (saved to your Pictures folder).
- Reading a file name: G3-Anon-**CTK**-**P1**T1-Unknown-**C**-0.avi = exercise CTK, person P1, **C = correct**. E1B1, E2B1, E3B1 = error E1, E2, E3.
- Left and right always mean **the person's left and right**, not the screen's.

Task 1. CTK: are E1 and E3 the same error? (4 videos)

Folder 1_CTK. One person (P1). In each video, take one screenshot at the moment **the arms are highest** (4 screenshots).

Video	Meaning	Name your screenshot
P1T1-...-C-0	Correct (same session as E1)	CTK_C_T1.png
P1T1-...-E1B1-0	Error E1	CTK_E1.png
P1T3-...-C-0	Correct (same session as E3)	CTK_C_T3.png
P1T3-...-E3B1-0	Error E3	CTK_E3.png

Question (circle one)	Answer
1-1. In E1, are the arms lower than in C?	yes / no / not sure
1-2. In E3, are the arms lower than in C?	yes / no / not sure
1-3. How is E3 different from E1? (tick all that apply)	(a) arm height (b) how close the arms come in front of the face (c) elbow bend (d) speed or pauses (e) only one arm (f) no difference (g) other: ____
1-4. In one sentence: what does E3 look like?	

Task 2. ELK: what is different about P2? (6 videos)

Folder 2_ELK. Three people (P1, P2, P3) × correct (C) and error E2 ("body is not tilted"). In each video, take one screenshot at the moment **the body leans most to the side** (6 screenshots). Name them like ELK_P1_C.png, ELK_P1_E2.png.

Question	P1	P2	P3
2-1. In E2, does the body clearly lean less than in C?	yes / no / not sure	yes / no / not sure	yes / no / not sure
2-2. How much does the body lean in C?	small / medium / big	small / medium / big	small / medium / big
2-3. Does the person face the camera straight on?	yes / no / not sure	yes / no / not sure	yes / no / not sure
2-4. Distance from the camera	near / normal / far	near / normal / far	near / normal / far
2-5. Anything else you notice (one line)			

What to send and when it is done

- **Two slides** (PowerPoint or Google Slides): slide 1 with the 4 CTK pictures side by side, slide 2 with the 6 ELK pictures; a short label under each (C, E1, E3, ...).
- **Your answers**: a photo of this sheet with your circles, or the same answers in a message.
- **Done when**: 10 pictures + answers to 1-1 to 1-4 and 2-1 to 2-5 ("not sure" is a good answer). Send to the team lead by Fri 16 Oct.

What you get out of it

- You make the **error example figure** for the presentation and the paper (you are credited).
- You learn by eye how physiotherapists judge exercise errors.
- **Optional (if you want)**: read page 2 of the teammate's report (KERAAL_Kinect_Model_Report.pdf), "The score: ROC-AUC", and explain it in three sentences in your own words. If you want more, run Step 1 of that report's code together with the team lead for 30 minutes (a first step in Python; every line is explained).

Please keep in mind

- The videos are for research only (CC BY-NC-SA). Do not post or forward them, and delete them when you finish. Keep the screenshots inside the team too.
- If you get stuck, ask right away. Asking is part of the job.